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Nanobubble technology in anaerobic digestion: A review

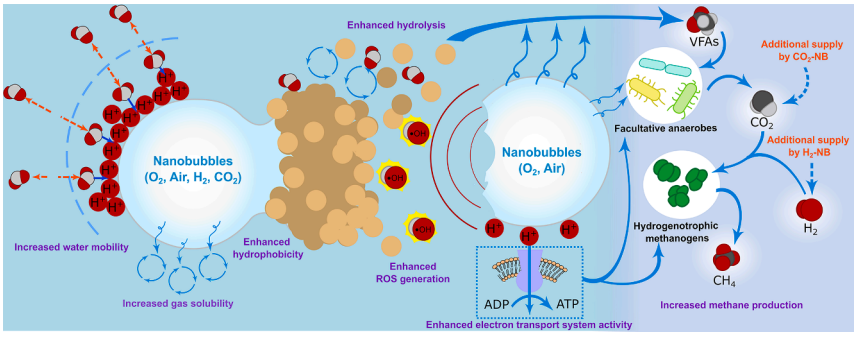
Wachiranon Chuenchart [a,1], Renisha Karki [b,1], Ty Shitanaka [b,1], Kyle Rafael Marcelino [a,1], Hui Lu [c], Samir Kumar Khanal [a,b,*]

[a] Department of Civil and Environmental Engineering, University of Hawai'i at Mānoa, 2540 Dole Street, Honolulu, HI 96822, United States
[b] Department of Molecular Biosciences and Bioengineering, University of Hawai'i at Mānoa, 1955 East-West Road, Honolulu, HI 96822, United States
[c] School of Environmental Science and Engineering, Sun Yat-sen University, Guangzhou, China

HIGHLIGHTS

- Nanobubble technology has potential to improve the anaerobic digestion process.
- Underlying mechanisms need to be fully elucidated.
- Development of process controls is required to facilitate new studies.
- In-situ nanobubble generation should be explored for industrial applications.
- Convergence of concepts in anaerobic digestion and nanobubble fields is needed.

GRAPHICAL ABSTRACT



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ABSTRACT

Nanobubble technology has significant potential to improve the anaerobic digestion (AD) process by ameliorating the rate-limiting steps of hydrolysis and methanogenesis, as well as providing process stability by reducing sulfide and volatile fatty acid (VFA) levels. Nanobubbles (NB) can enhance substrate accessibility, digestibility, and enzymatic activity due to their minuscule size, high electrostatic interaction, and ability to generate reactive oxygen species. Air- and O_2-NB can create a microaerobic environment for higher efficiency of the electron transport system, thereby reducing VFAs through enhanced facultative bacterial activity. Additionally, H_2- and CO_2-NB can improve hydrogenotrophic methanogenesis. Recently, several studies have employed NB technology in the AD process. There is, however, a lack of concise, synthesized information on NB applications to the AD process. This review provides an in-depth discussion on the NB-integrated AD process and the putative mechanisms involved. General discussions on other potential applications and future research directions are also provided.

* Corresponding author at: Department of Molecular Biosciences and Bioengineering, University of Hawai'i at Mānoa, 1955 East-West Road, Honolulu, HI 96822, United States.
E-mail address: khanal@hawaii.edu (S.K. Khanal).
[1] The authors contributed equally to this work.

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1. Introduction

Anaerobic digestion (AD) is a microbial-mediated mature technology widely employed for waste remediation with concomitant generation of renewable energy. Depending on the feedstocks used, AD has been successfully adopted to produce various value-added biobased products such as organic fertilizers, short and medium chain volatile fatty acids (VFAs), and biopolymers (Sawatdeenarunat et al., 2016). In addition, AD has significant potential in curtailing greenhouse gas emissions as an environmentally sustainable technology (Surendra et al., 2014). However, AD also suffers from several drawbacks, such as a slow hydrolysis rate, especially for complex substrates (Shrestha et al., 2017); generation of VFAs from rapidly biodegradable substrates such as food waste (Chuenchart et al., 2020); generation of inhibitory products such as ammonia and hydrogen sulfide (H_2S/HS^-); and high susceptibility to changes in environmental factors, among others (Khanal and Li, 2017).

Some of these demerits can be addressed by adopting various pretreatments (Zhang et al., 2016; Shrestha et al., 2017). However, high energy and chemical requirements, coupled with the generation of secondary pollutants, could be cost-prohibitive at an industrial scale. Microaeration has emerged as a promising technology to address some of these drawbacks (Nguyen and Khanal, 2018). In microaeration, small quantities of air or O_2 are introduced in an anaerobic digester, which provides an environment conducive for the growth of facultative hydrolytic and acidogenic microorganisms. However, because microaeration demands effective gas transfer and precise air/O_2 dosing control, this could add complexity to the AD process. Moreover, there are also potential risks of explosions due to the rapid escape of airO_2 in biogas during microaeration. Therefore, alternative strategies are needed to overcome the above-stated lacuna.

One of the innovative strategies currently being explored in conjunction with AD is the application of nanobubble technology, an emerging gas delivery technology that produces bubbles with diameters 50–200 nm. In studies so far, nanobubble waters with different types of gases have been generated and used as an additive for AD batch systems. However, an interesting alternative to this is to generate nanobubbles (NB) in-situ through the use of membranes, lending additional flexibility to the design and operation of AD systems. The unique characteristics of NB can impact the physicochemical properties of solutions through enhanced gas solubility and high electrostatic interactions, among others (Atkinson et al., 2019). Substrate digestibility can be improved by the generation of reactive oxygen species (ROS), which can enhance the oxidation of organic matters (Wang et al., 2020a). In addition, NB can induce microaerobic conditions via air- and O_2-NB, resulting in improved performance and stability in the AD process through increased facultative bacterial activity and hydrogenotrophic methanogenesis (Nguyen and Khanal, 2018, Nguyen et al., 2019). NB have several potential applications in various fields, such as in environmental remediation (Barry et al., 2014; Zheng et al., 2015), biomedical engineering (Cai et al., 2015; Teirlinck et al., 2018), aquaculture and aquaponics (Jhunkeaw et al., 2021; Rahmawati et al., 2020; Wongkiew et al., 2021), and agriculture (Liu et al., 2013, 2016a, 2016b).

This review critically discusses recent advances in the state-of-the-art of NB-integrated AD processes, the potential underlying mechanisms associated with improved performance, and concludes with future research directions.

2. Brief overview of nanobubbles and potential mechanisms in anaerobic digestion

2.1. Overview of nanobubbles

NB (also known as ultrafine bubbles) are gaseous spherical bubbles with a diameter less than 1 μm, typically ranging between 50 and 200 nm (ISO 20480–1:2017). The most common generation method, hydrodynamic cavitation, generates NB via a gas-fed pressurized chamber when discharged at atmospheric pressure (Panda et al., 2020; Zhou et al., 2019). Since the mid-1990s, the existence of NB was debated by researchers, citing that bubbles of such small magnitudes have no thermodynamic basis for existence as per the Epstein and Plesset Theory of Bubble Dissolution and Growth (Epstein and Plesset, 1950) and the Laplace Pressure Bubble Catastrophe effect (Alheshibri et al., 2016). However, with the publication of the first images of NB in 2000 (Lou et al., 2000), their existence has increasingly been accepted and established with studies characterizing their unique physicochemical properties such as ROS generation, low buoyancy, long residence time, high surface charge, high negative zeta potential, and enhanced gas solubility (Lyu et al., 2019).

NB are stable in solution with residence times of days to months. Unlike bubbles of 0.1 μm or greater, NB are not affected by buoyancy forces (which leads to the separation of a bubble out of solution) and are mainly governed by Brownian motion (Nirmalkar et al., 2018). NB are also stable due to a high electrostatic interaction associated with a strong negative surface charge and zeta potential. The zeta potential is a measure of the electrical potential around the interface of NB. High negative zeta potential is caused by the adsorption of hydroxide ions at the interfacial surface of NB, resulting in an electrical double layer of ions and counter ions (Takahashi, 2005). Both the surface charge and the zeta potential are important attributes that facilitate higher cation mass transfer, hydrophobic forces, and water mobility.

The generation of ROS (e.g., hydrogen peroxide (H_2O_2), hydroxyl (•OH), superoxide (O_2•-), or singlet oxygen ($^1O_2$)) upon collapse is one of the key features of chemically reactive NB (Atkinson et al., 2019). Ahmed et al. (2018) reported that the dominant ROS in NB is •OH, with significant amounts being generated by air- and O_2-NB. Thus, •OH derived specifically from O_3-, O_2-, or air-NB has significant potential to oxidize complex organic matters, nutrients, pathogens, and other emerging contaminants. In the wastewater treatment process, •OH is generated using O_3, UV irradiation, and other advanced oxidation processes (Atkinson et al., 2019). However, these processes are energy and chemically intensive (Atkinson et al., 2019). Therefore, NB can be an innovative alternative to lower chemical and energy usage.

2.2. Potential mechanisms of nanobubbles in anaerobic digestion

NB technology can improve the performance and efficiency of the AD process, explicitly enhancing the rate-limiting steps: hydrolysis and methanogenesis. Application of NB in AD process is at a nascent stage, and there is significant potential to explore different gas types (air, CO_2, O_2, and H_2) on several feedstocks such as animal manure, sewage sludge, food waste, lignocellulosic biomass, and various complex industrial wastewaters. The mechanisms involved regarding NB in improving the AD performance are not well-delineated; however, potential inferences are discussed in Section 3.

Each NB gas species has unique attributes, which can be applied at different steps in the AD process. Wang et al. (2020b) demonstrated increased electron transport system (ETS) activity in AD through air- and O_2-NB by the promotion of higher adenosine triphosphate (ATP) synthesis and metabolite transport. Additionally, CO_2- and H_2-NB can act as substrates to enhance hydrogenotrophic methanogenesis, with molecular H_2 acting as an electron donor (Goldman et al., 2009).

NB disturb the hydrogen-bonding network of water (as evidenced from higher spin–spin relaxation time), thereby increasing water mobility (Liu et al., 2013), allowing for higher hydration of organic particles, and improving the penetration of substrates into inner regions of biofilms. Research on water mobility in AD is limited, but studies have observed a positive correlation between hydrolysis rates and water mobility (Hong et al., 2016; Yang et al., 2019). This increased water mobility is explained by the high zeta potential of NB (Takahashi, 2005). Furthermore, other NB stability studies have shown that a higher zeta potential correlates to higher water mobility between different NB gas species ($H_2 > CO_2 \approx$ air) (Fan et al., 2020a; Wang et al., 2020c).

Substrate digestibility and accessibility in AD are enhanced via ROS activity and high electrostatic interactions, respectively, in NB that govern both cation mass transfer (enhanced localized nutrient uptake) and hydrophobicity (enhanced particle attachment). The details of these mechanisms are merely inferred and not experimentally examined.

3. Effects of nanobubbles on anaerobic digestion processes

3.1. Performance and efficiency

The anaerobic digestibility of synthetic substrates (cellulose and lignin) and biomass (corn straw, food waste, waste activated sludge and pig manure) was evaluated in batch systems (via biomethane potential (BMP) tests) with NB water. Methane productivities of NB with different gases showed improvements between 3−40% relative to a control group (addition of an equivalent volume of deionized water (DW)), as shown in Table 1. Additionally, the substrate biodegradation rate was governed by the NB gas species, volume of NB water added, type of substrates used, and operating conditions of the batch reactor.

NB have been shown to enhance the biodegradation of cellulosic substrates due to ROS activity, increased water mobility, and improved ETS activity, as discussed earlier in Section 2.2. Air-NB promoted a microaerobic environment in AD, resulting in an increase in abundance of lignocellulose-degrading microbes and higher amounts of cellulose reduction compared to CO_2-NB (Wang et al., 2020b). A follow-up study by the same authors found that increasing O_2 concentrations led to higher methane yields with mixed O_2-NB:N_2-NB at ratios of 1:4, 1:1 and 3:1 (Wang et al., 2020a). Although there were inconsistencies between the studies, the highest levels of both cellulose crystallinity reduction (%) and ETS activity ($mg{\cdot}g^{-1}{\cdot}h^{-1}$) were found with O_2-NB as compared to other NB gas species (Wang et al., 2020a).

A study conducted by Ho et al. (2020) found that NB enhanced the biodegradation of refractory lignin by 19.3% (N_2-NB), followed by 17.4% (CO_2-NB), and 6.4% (control). This resulted in a 22% (N_2-NB) and 17% (CO_2-NB) increase in methane yield over the control. The generation of •OH from NB may explain this result, which can contribute to C–C or C–O bond cleavage within lignin molecules (Yang et al., 2020).

Wang et al. (2020a) investigated microbial adaptation using O_2-NB and N_2-NB in the pre-augmentation of corn straw. The highest reduction in cellulose content was observed by using O_2-NB (38%) followed by N_2-NB (33%). Additionally, reductions in hemicellulose and lignin were similar for O_2-NB and N_2-NB (35% and 9%, respectively). However, methane yield was significantly different for O_2-NB and N_2-NB (22.3% and 9.8%, respectively), suggesting that the microaerobic environment greatly promoted cellulose hydrolysis in lignocellulose degradation.

Hou et al. (2021) observed the effect of N_2- and air-NB in two-stage food waste AD. In the first stage (hydrolysis and acidogenesis), N_2- and air-NB led to higher H_2 production of approximately 23.7% and 39.9%, respectively, compared to control (DW). In the second stage, N_2- and air-NB increased methane production by 15.2% and 24.7%, respectively, relative to DW. Additionally, protein degradation, glucose breakdown, and enzyme activity followed the previous trend of biogas yield of air-NB > N_2-NB > DW. These results indicate that N_2-NB primarily enhanced hydrolysis/acidification, whereas air-NB improved hydrolysis/acidification and methanogenesis.

Furthermore, NB were found to enhance hydrolysis of organic substrates to generate higher amounts of VFAs in early stages (3–12 days) (Fan et al., 2020a; 2020b; Yang et al., 2020; Yang et al., 2019). However, final VFA concentrations were inconsistent. VFA concentrations were relatively low for waste activated sludge (Yang et al., 2020) and pig manure (Fan et al., 2020b), but higher for food waste (Hou et al., 2021) and cellulose (Wang et al., 2020b) compared to DW. In high-solid pig manure AD, adding CO_2-NB was able to reduce VFA accumulation at 8% total solids (TS), but did not reduce VFAs at 10% TS. This was because total VFAs in the 10% TS trial were higher than the inhibitory threshold of approximately 8700 mg acetic acid equivalent (HAc)·L^{-1} (Fan et al., 2020b), which suggests that benefits of NB may be limited. Future research should focus on using different gas species in NB generation (e.g., H_2 and CO_2) to better understand NB effects on AD of diverse feedstocks. Furthermore, specific NB gas species can be added at different times to observe NB effects on each stage (e.g., hydrolysis/acidogenesis and methanogenesis) of two-stage AD process.

3.2. Enzyme activity

As stated in Section 2.2, NB may increase the water mobility of the surrounding aqueous environment resulting in enhanced biological activity. Through increased water mobility, there is a higher mass transfer of nutrients to microbial cells, which improves enzymatic activity, resulting in higher decomposition rates of substrates with more biogas production (Wang et al., 2019). Air-NB was reported to enhance activities of extracellular hydrolases such as alkaline phosphatase (ALP), acid phosphatase (ACP), α-glucosidase, and protease by 29%, 47%, 14%, and 8%, respectively (Hou et al., 2021) compared to DW. These enzymes degrade macromolecules such as proteins and polysaccharides thereby facilitating faster conversion into VFAs, which are then utilized by methanogens to generate methane (Yang et al., 2019).

Coenzyme F_{420} is an essential hydride carrier in hydrogenotrophic methanogenesis (Lambrecht et al., 2017). Hou et al. (2021) reported a 140% increase in the relative activity of coenzyme F_{420} in air-NB reactors compared to DW reactors. Additionally, there was a 34% increase in coenzyme F_{420} activity in air-NB reactors over N_2-NB reactors. Moreover, the air-NB reactors had 30% higher cellulase activities than DW reactors (Wang et al., 2020b). Furthermore, another study showed that N_2-NB addition to waste activated sludge improved the enzyme activity by 14–17% compared to DW (Yang et al., 2019). However, due to inconsistencies in AD operating conditions, it is still unclear of the role of enzymes on methane production.

3.3. Microbial community and activity

Higher relative abundances of *Chloroflexi*, *Bacteroidetes*, *Spirochaete*, *Synergistetes,* and *Firmicutes* were observed in air-NB reactors than DW reactors during the degradation of cellulosic biomass (Wang et al., 2020b, 2020c). The phyla *Firmicutes* and *Bacteroidetes* are mainly associated with degrading lignocellulosic biomass (Shrestha et al., 2017). The relative abundance of *Bacteroidetes* was significantly higher in the air-NB reactor than the DW reactor, which may have caused an increase in cellulose degrading efficiency (Wang et al., 2020b). However, in a follow-up study by Wang et al. (2020c), the relative abundance of *Bacteroidetes* in NB reactors was only slightly higher than in the DW reactor. Additionally, the relative abundances of *Methanosaeta* in the air-NB (73.4%) and CO_2-NB (52.7%) reactors were 1.64-fold and 1.18-fold higher, respectively, than that in DW reactors (44.8%) (Wang et al., 2020c). The relative abundance of *Methanosaeta* followed a decreasing order of O_2-NB (70.7%) > N_2/O_2 (1:3)-NB (65.5%) > N_2-NB (64.6%) > N_2/O_2 (4:1)-NB (60.7%) > air-NB (55.1%) > DW (37.4%) > N_2/O_2 (1:1)-NB (34.8%) (Wang et al., 2020b).

The enrichment of *Methanomassiliicoccus* in NB reactors can enhance methane production by utilizing H_2 and methanol or methylamines as substrates (Wang et al., 2020b). Moreover, it was found that air-NB or O_2-NB reactors had the highest relative abundance of *Methanobacterium*, which can enhance interspecies H_2 transfer (Wang et al., 2020b, 2020c). Additionally, O_2-NB, CO_2-NB, and N_2-NB promoted the relative abundance of *Methanolinea*, which utilizes CO_2 to produce methane. Similar results were observed in a study by Nguyen et al. (2019), where intermittent microaeration (at an oxidation-reduction potential (ORP) of −470 mV) resulted in a significant increase in the relative abundance of facultative bacteria (2-fold increase of the genus *Proteiniphilum*).

Studies linking AD performance with the relative abundances of the microbial community to different NB gas species are very limited.

Table 1
NB-integrated AD systems under mesophilic conditions.

Substrate	Gas	NB: total volume	Condition		SMY ($mL \cdot g^{-1}$ $VS_{reduced}$)			VS reduction (%)			Remarks	Reference
					DW	NB	% compared with DW	DW	NB	% compared with DW		
WAS (14 days) S/I = 1VS basis	N_2	100 g/ 200 g	N_2-NB		311.4	402.1	29.1	37.5	38.0	1.3	Reductions of protein and carbohydrate were 47.6 and 262.0%, respectively VFAs and TOC both increased by 31.4% on day 3 Specific activities of ALP, ACP, α-Glu, and Pro were increased by 16.9, 14.1, 15.8, and 17.1%, respectively (compared to control)	(Yang et al., 2019)
WASS/I = 2 (20 days) VS basis	N_2	100 mL/ 200 mL	N_2-NB (mL) with lignin50 $mg \cdot L^{-1}$	25	378.4	389.7	3.0	22.9	25.6	11.8		(Yang et al., 2020)
				50		403.5	6.6		27.5	20.1		
				100		443.1	17.1		30.3	32.3		
			Lignin $mg \cdot L^{-1}$ 100% NB	50	364.6	441.3	21.0	NA			Degradations of lignin were 53.9, 64.4, 67.1, 70.3, and 61.6%, with respect to increase in lignin concentration	
				100		457.9	25.6					
				200		478.6	31.3					
				300		475.1	30.3					
FW (68 hr and 24 days) S/I = 6.5 and 4.5 (VS basis)	N_2 and air	70 mL/ 150 mL	First stage	N_2–NB	19.8 (H_2)	24.5	23.7	NA			Reductions of protein on day 8 were 67, 79, and 83% for DW, N_2 and air-NB, respectively Reduction of glucose on day 12 were 69, 76, and 83%, respectively. Specific activity of α-Glu was significantly increased by 23% in N2-NB, while air-NB enhanced specific activity of ACP by 47% compared to control	(Hou et al., 2021)
				Air-NB		27.7	39.9					
		20 mL/ 150 mL	Second stage	N_2–NB	300.9VS added	346.6	15.2	NA			Increase in specific activity of co-enzyme F_{420} for N_2-NB (4%) and air-NB (34%) compared to control Final VS degradation of 66.0, 71.2, 74.3%, for DW, N_2-NB and air-NB, respectively	
				Air-NB		375.3	24.7					
Corn straw (24 days) S/I = 1.5	N_2 and O_2	200 mL/ 400 mL	NB adding in 6 times pre-augmented for 216 days	N_2–NB	115.9	127.3	9.8	NA			Cellulose reduction was the highest in O_2-NB (38%) compared to control Similar hemicellulose, lignin, and crystallinity reductions between N_2-NB and O_2-NB Activity of ETS_{max}: 134.8, 149.1, and 155.2 $mg \cdot g^{-1} \cdot h^{-1}$ for DW, N_2– NB, and O_2-NB, respectively	(Wang et al., 2020a)
				O_2-NB		141.7	22.3					
Cellulose (50 days) S/I = 3.5 (VS basis)	CO_2andair	30 mL/ 60 mL	NB	CO_2-NB	223.3	246.0	10.2	44.7	53.6	19.9	Reduction of crystallinity: 67.4, 76.9, and 43.1%, for DW, CO_2-NB and air-NB, respectively Reductions of cellulose with CO_2-NB and air-NB were 21 and 29% higher	(Wang et al., 2020c)
				Air-NB		263.6	18.0	NA	58.0	29.8		

(*continued on next page*)

Table 1 (*continued*)

Substrate	Gas	NB: total volume	Condition		SMY ($mL \cdot g^{-1}$ $VS_{reduced}$) DW	NB	% compared with DW	VS reduction (%) DW	NB	% compared with DW	Remarks	Reference
											than control, respectively Reductions of TOC were 38.6, 39.6, are 43.1%, for DW, CO_2-NB and air-NB, respectively	
HAc:Lignin 90%:10% (TOC basis) 2 g TOC·L^{-1} (10 days)	N_2 and CO_2	NA	NB	CO_2-NB	599TOC removed	703	17.2	NA			Reductions of lignin were 6.4, 17.4, and 19.3% for DW, CO_2-NB, and N_2-NB, respectively The highest lignin reduction (42.7%) with N_2-NB was observed for the ratio of 95.5:5 HAc: lignin (TOC basis)	(Ho et al., 2020)
				N_2-NB		728	21.5					
CelluloseS/I = 1 (VS basis)	N_2, air, and O_2	35 mL/ 70 mL	NB	Air-NB	179.4	193.3	7.7	NA			Reductions of cellulose were 84.9, 92.1, 94.8, 96.5% for DW, air-NB, N_2-NB, and O_2-NB, respectively Reductions of cellulose crystallinity were 31.9, 36.0, 37.5, and 38.6%, for DW, air-NB, N_2-NB, and O_2-NB, respectively ETS of samples: $O_2 > N_2 >$ air > DW	(Wang et al., 2020b)
				N_2-NB		196.1	9.3					
				O_2-NB		233.0	29.9					
Swine manure (28 days) S/I = 0.25	H_2, CO_2, andair	100 g/ 200 g	NB	Air-NB	≈ 161	192	19.1	NA			Activities of ALP, ACP, α-Glu, Pro, dehydrogenase, co-enzyme F_{420}, and ETS in NB are higher than the control	(Fan et al., 2020a)
				CO_2-NB		195	20.8					
				H_2-NB		225	39.3					
Pig manure (68 days)	CO_2	100 g/ 200 g	TS (%)	3	NA	201	20.3	NA			Reactor (8% TS) DW and both reactor (10% TS) DW and (10% TS) NB were inhibited with pH around 6.3 and FAN around 50 mg·N L^{-1}	(Fan et al., 2020b)
				6	NA	230	25.0					
				8	24fail	261	996.6					
				10	fail	fail	NA					
WAS (30 days) S/I = 0.2	N_2, air, CO_2, and H_2	100 mL/ 200 mL	Stepwise NB water addition 25 mL on day 0, 4, 8, and 12	N_2-NB	190.9VS added	211.6	10.8	NA			Addition of NB led to 10–17% higher SMY compared to control	(Wang et al., 2019)
				Air-NB		218.2	14.3					
				CO_2-NB		222.4	16.5					
				H_2-NB		223.9	17.3					

DW: Deionized water; NB: Nanobubbles; ALP: Alkaline phosphatase; ACP: Acid phosphatase; α-Glu: α-glucosidase; VFA: Volatile fatty acids; TOC: Total organic carbon; ETS: Electron transport system; FAN: Free ammonia nitrogen; FW: Food waste; HAc: Acetic acid; HBu: Butyric acid; ORP: Oxidation-reduction potential; Pro: Protease; VS: Volatile solids; S/I: Substrate-to-inoculum ratio; SMY: Specific methane yield; TS: Total solids; WAS: Waste activated sludge; NA: Not available

Therefore, further research is needed to better understand the effects of NB on the microbial community in AD. Approaches that link enzyme activity with specific microbial communities need to be examined to gain insight into how NB influence microbial activity.

4. Potential applications of nanobubble technology

4.1. Application of different nanobubble gas species

Different NB gas species such as air, N_2, CO_2, O_2, or H_2 (Fan et al., 2020a, Fan et al., 2020b; Hou et al., 2021; Wang et al., 2019; Wang et al., 2020a; Wang et al., 2020b; Wang et al., 2020c; Yang et al., 2019) were employed to enhance the degradation of various substrates to improve biogas production. However, each NB gas has unique physicochemical properties, and studies on the effects of different gas combinations and mixture ratios in the AD process are scarce. Further research on this very subject is needed to realize the full potential for enhanced degradation of diverse substrates, digestibility, and overall biogas production.

Some gaseous species such as H_2 and CO_2 can be supplied for hydrogenotrophic methanogenesis, especially for biological biogas upgrading, as discussed in Section 2.2. Biogas upgrading by H_2-NB may increase methane concentration due to higher hydrogenotrophic methanogenic activity (Bassani et al., 2015; Fan et al., 2020; Wang et al., 2019). However, the effect of adding H_2- and CO_2-NB to enhance hydrogenotrophic methanogenesis still needs to be examined. Using air-

or O_2-NB for microaeration can support ETS activity, hydrolysis, and hydrogenotrophic methanogenesis. Furthermore, operational issues such as sulfide toxicity (especially to methanogens) and pipe and reactor corrosion due to high H_2S/HS^- level can be overcome by supplying air- or O_2-NB into the AD system. For example, when microaeration (with O_2) during AD of energy crops (biomass) was employed, Nguyen et al. (2019) reported better process stability due to increased activity of facultative anaerobes that rapidly scavenged accumulated VFAs in the reactor. In another study, Khanal and Huang (2003) successfully alleviated sulfide toxicity to methanogens during anaerobic treatment of high sulfate wastewater via microaeration. Therefore, adding air- or O_2-NB into the AD reactors may increase reaction rates compared to conventional microaeration due to higher surface area, gas transfer efficiency, and increased ETS activity. However, further studies are needed to elucidate the underlying mechanisms involved.

4.2. Pretreatment with nanobubbles

NB can also be used as a pretreatment method for highly recalcitrant substrates such as lignocellulosic biomass. Nguyen et al. (2019) employed microaeration during AD of complex biomass substrates and reported enhanced hydrolysis. However, more studies are needed to examine how supplementation of different NB gas species affects hydrolysis. It has been widely accepted that the degradation of lignin requires air or O_2 for enzymatic cleavage of aromatic rings in lignin (Shrestha et al., 2017). Thus, air- or O_2-NB may enhance lignin degradation. Additional research should be conducted to gain insight into lignin degradation with the use of air- or O_2-NB.

4.3. Defouling in anaerobic membrane bioreactors

Membrane fouling is a critically important factor affecting anaerobic membrane bioreactor (AnMBR) performance, resulting in membrane flux reduction and increased operation costs associated with high chemical/energy usage and frequent membrane replacement (Lin et al., 2013). One alternative to alleviate this issue is to employ NB technology in generating •OH for the removal/detachment of biofilms from the membrane surfaces (Agarwal et al., 2011). The use of ultrasonication as a mechanical stimulus can be used in conjunction with intermittent NB generation to effectively clean membranes (Broekman et al., 2010). Extracellular polymeric substances (e.g., proteins and polysaccharides) produced by the microbes can irreversibly bind to the membranes with structural enhancement from cations such as calcium and magnesium (Horseman et al., 2021). NB could also help to prevent the formation of strongly bonded biofilms by removing the above-stated cations from the solution due to their highly negative gas–liquid interface (Dayarathne et al., 2019).

Other benefits of NB in AnMBR include reducing the use of harmful chemicals (e.g., sodium hypochlorite, nitric acid, ethylenediaminetetraacetic acid), thereby reducing the cost through increasing the lifespan of the membranes due to less intensive cleaning processes. However, •OH needs to be precisely controlled, as high concentration may negatively affect the microbial communities involved. Further research is needed to examine •OH formation when NB generation is coupled with ultrasonication and its subsequent effect on AnMBR fouling control.

5. Engineering implications

In-situ NB generation in AD processes can certainly augment current microaeration and biogas upgrading strategies, but several engineering design considerations must be carefully accounted for. Similarly, for microaerated anaerobic systems, the integration of ORP-based process controls (i.e., the usage of sensors, controllers, and actuators) will be essential. Additionally, other process parameters such as aeration frequency, gas flow rate, oxygen transfer rate, and oxygen utilization rate must be included (Chen et al., 2020; Nguyen et al., 2015).

NB generators utilizing hydrodynamic cavitation could potentially bring a slew of problems for laboratory and field-scale reactors, making this method extremely difficult if NB are not pre-generated in a separate compartment within the reactor. The high fluid flow rate required for hydrodynamic cavitation (Oliveira et al., 2018) could disturb and negatively impact the reactor content and the associated microbial communities. There are, however, several simple methods of generating NB, e.g., via ultrasonication (Cho et al., 2005), electrochemistry (Ranaweera and Luo, 2020), and most interestingly, via porous membranes (Kukizaki and Goto, 2006). For AD, in-situ NB generation through a submerged membrane appears to be the most attractive option due to its simplicity in requiring only gas flow through a microaeration process control unit.

Thus far, there are no studies or field applications of NB-microaeration using membrane in-situ NB generation. However, based on data from other derivative studies, the development of these technologies can bring further improvement to the AD process. For example, membrane-aerated biofilm reactors have been used to treat effluent from AD (Camiloti et al., 2019; Chen et al., 2015; Gong et al., 2020; Peng et al., 2020). The aerated membrane supported the attachment of biofilms harboring specific microbes that enhanced sulfide (sulfide oxidizing bacteria) and ammonia reduction (anammox and denitrifying anaerobic methane oxidation), among others.

6. Perspectives

NB technology provides new opportunities in AD due to their minuscule size, inherent microaeration ability (with gas containing air or O_2), ROS generation, high zeta potential, high surface charge, and high gas transfer efficiency. However, there are many challenges to overcome for the widespread deployment of this technology. Because of the lack of mechanistic understanding and delivery methods for NB-integrated AD systems, further research is needed to facilitate new opportunities.

Fundamental NB research needs to be well-established with stability assessments focusing on the zeta potential, residence time, spin–spin relaxation time, and bubble size/concentration in various complex aqueous matrices with different NB gas species. Moreover, studies on NB-derived ROS need to be fully delineated as their existence and impact on substrates and biofilms are not well examined.

NB-integrated AD systems are at an early stage of research and development, with most studies limited to batch systems. These systems do not provide reliable information to evaluate the efficacy of NB technology for different industrial-scale applications in AD; thus, future research must expand to new avenues through continuous studies. Continuous systems coupled with versatile and scalable NB systems that utilize aeration/oxygenation process control systems should be developed. The development of such systems can allow for ease-of-use and afford new explorations as to how different NB gas species can be delivered in the AD process. Future studies should explore various process parameters (organic loading rate, hydraulic retention time/solids retention time); substrate characteristics (total solids, volatile solids, and total organic carbon); and the microbial communities involved during long-term operations. Future studies should also explore the removal of micropollutants, pathogens, and antibiotic resistant genes and antibiotic resistant bacteria through air- and O_2-NB.

7. Conclusions

There is growing research interest in applying NB technology in the AD processes. NB of various gas species can enhance the hydrolysis of complex substrates, thereby improving overall digestibility. Air- and O_2-NB can enhance sulfide control, system stability, and biogas production of AD systems through increased activities of sulfide-oxidizing bacteria, facultative anaerobes, and methanogens. These effects can be ascribed

to increased water mobility, perpetual ROS generation, and enhanced microbial and ETS activity depending on the NB gas species. With improvements in delivery methods and deeper integration of both research areas, NB technology can provide new opportunities in the AD field.

CRediT authorship contribution statement

Wachiranon Chuenchart: Investigation, Visualization, Formal analysis, Writing - original draft. **Renisha Karki:** Investigation, Visualization, Formal analysis, Writing - original draft. **Ty Shitanaka:** Investigation, Visualization, Formal analysis, Writing - original draft. **Kyle Rafael Marcelino:** Investigation, Visualization, Formal analysis, Writing - original draft. **Hui Lu:** Writing - review & editing. **Samir Kumar Khanal:** Supervision, Conceptualization, Writing - review & editing.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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